

SONIQ

Employee Productivity & Intelligence Platform

BUSINESS REQUIREMENTS DOCUMENT

Phase 1 — Frozen Scope | Version 3.0 | April 2026

◆ **STATUS: FROZEN — APPROVED FOR DEVELOPMENT** ◆

Property	Details
Product Name	SonIQ — Employee Productivity & Intelligence Platform
Document Version	3.0 (Complete Revised BRD)
Status	FROZEN — Approved for Full Development
Target Completion	End of May 2026 (including full testing)
Date	April 2026
Data Engine	Trackpilots EXE + Webhook Integration (3 webhook events)
UI Design Language	SonIQ Design System (inspired by Effortly — fully unique identity)
Panel Architecture	Two Separate Panels: Super Admin Portal + Client Portal
Font	Poppins
Intended Audience	Frontend Devs, Backend Devs, QA Team, Product Owner, Stakeholders

1. Executive Summary

1.1 Product Vision

SonIQ is a B2B SaaS Employee Productivity & Intelligence Platform. It ingests real-time workforce tracking data from Trackpilots desktop agent via webhooks, processes and stores that data on Soniq's own infrastructure, and presents it through a high-quality, uniquely designed intelligence dashboard. Soniq is not a clone of Trackpilots or Effortly — it is a distinct product with its own identity, naming, design system, and user flows.

1.2 Panel Architecture — TWO Separate Portals

CRITICAL ARCHITECTURE DECISION

SonIQ has TWO completely separate web portals: (1) SUPER ADMIN PORTAL — Internal Soniq team only. URL: admin.soniq.io — Manages all client orgs, billing, platform health, global settings. (2) CLIENT PORTAL — For each client organization's Admin and Manager. URL: app.soniq.io — Manages their org's employees, teams, reports, settings. These are NOT role-based tabs in one app. They are two independent frontend applications with separate codebases, separate auth, and separate APIs.

Portal	URL	Users	Purpose
Super Admin Portal	admin.soniq.io	SonIQ Internal Team only	Manage all client orgs, billing, platform monitoring, global config
Client Portal	app.soniq.io	Client Admin + Client Manager	Manage their org's productivity data, teams, reports, settings

1.3 Data Flow Pipeline (Frozen Architecture)

HOW SONIQ WORKS — STEP BY STEP

Step 1: Trackpilots EXE is installed on employee machines (managed by Trackpilots). Step 2: EXE captures activity, app/website usage, and screenshots in real time. Step 3: Trackpilots pushes 3 webhook event types to Soniq's registered HTTPS endpoint with HMAC SHA256 signature. Step 4: Soniq validates the signature, processes the payload, stores to PostgreSQL DB + images to S3. Step 5: Soniq's internal REST APIs serve processed data to Client Portal and Super Admin Portal. Step 6 (Future): Soniq builds its own proprietary EXE agent to replace the Trackpilots dependency entirely.

1.4 Three Trackpilots Webhook Events

#	Event Name	Trigger	Key Data Fields
1	desktop.activity_tracking.captured	User work mode or privacy mode changes	organisationId, teamId, userId, workMode (boolean), timestamp
2	desktop.app_tracking.captured	App/website tracked on employee desktop	organisationId, teamId, userId, app.name, app.type, app.category, app.domain, app.fullUrl, app.productivityStatus, time.startDate, time.endDate, time.durationInSeconds, trackingMode, operatingSystem
3	desktop.screenshot_tracking.captured	Screenshot captured at configured interval	organisationId, teamId, userId, screenshotId, imageBuffer (base64 JPEG up to 50MB), app.name, app.domain, app.productivityStatus, time.capturedAt, operatingSystem, workType, isIdle

1.5 User Roles Summary

Role	Portal	Scope	Access Level
Super Admin	Super Admin Portal (admin.soniq.io)	Entire platform — all clients, all data	Full platform control — create/suspend orgs, billing, audit, global settings
Client Admin	Client Portal (app.soniq.io)	Own organization only — all teams	Full org visibility — dashboard, teams, reports, screenshots, settings, roles
Client Manager	Client Portal (app.soniq.io)	Assigned teams only — read access	Limited to their team's data — cannot access settings or other teams
Employee	No portal login (Phase 1)	Appears as data subject only	Monitored via EXE — self-service portal is out of scope for Phase 1

2. Pre-Development Dependencies & Setup

Before a single line of Soniq code is written, the following infrastructure, accounts, and agreements must be in place. This is the complete pre-development checklist.

2.1 Infrastructure to Purchase / Set Up

#	Item	Service	Purpose	Priority
1	Cloud Server (Production)	AWS EC2 / DigitalOcean / Hetzner (min 4 vCPU, 8GB RAM, 100GB SSD)	Host Soniq backend API, webhook receiver, queue processor	Critical – Day 1
2	Cloud Server (Staging)	Same provider, smaller instance (2 vCPU, 4GB RAM)	Staging/testing environment that mirrors production	Critical – Day 1
3	Object Storage (Screenshots)	AWS S3 / DigitalOcean Spaces / Cloudflare R2	Store all screenshot images received from Trackpilots webhooks	Critical – Day 1
4	Managed Database	AWS RDS PostgreSQL / Supabase / DigitalOcean Managed DB (PostgreSQL 15+)	Primary relational database for all Soniq data	Critical – Day 1
5	Cache / Queue	Redis Cloud / Upstash Redis	Webhook event queue (BullMQ), dashboard data cache, live status cache	Critical – Day 1
6	CDN	Cloudflare CDN (free tier sufficient for Phase 1)	Serve screenshot images fast globally; DDoS protection; SSL termination	Critical – Day 1
7	Domain Names	Domain registrar (GoDaddy / Namecheap / Cloudflare)	soniq.io (or chosen domain), app.soniq.io, admin.soniq.io, api.soniq.io	Critical – Day 1
8	SSL Certificates	Cloudflare / Let's Encrypt (auto via Cloudflare)	HTTPS for all subdomains – required for Trackpilots webhook delivery	Critical – Day 1
9	Transactional Email	Resend / SendGrid / AWS SES	Send invite emails, notifications, password resets to client users	Critical – Week 1

#	Item	Service	Purpose	Priority
10	Error Monitoring	Sentry (free tier available)	Capture and alert on backend errors, webhook processing failures	Important — Week 1
11	Log Management	Logtail / Papertrail / AWS CloudWatch	Centralized logging for webhook events, API calls, errors	Important — Week 1
12	CI/CD Pipeline	GitHub Actions (free) / GitLab CI	Automated testing and deployment to staging and production	Important — Week 2
13	Secret Management	AWS Secrets Manager / HashiCorp Vault / Doppler	Store Trackpilots API keys, webhook secrets, DB credentials securely	Critical — Week 1
14	Backup Storage	S3 bucket (separate from screenshots)	Automated daily DB backups and screenshot backups	Important — Week 2

2.2 What to Ask / Get from Trackpilots (Dependency Checklist)

TRACKPILOTS DEPENDENCY — MUST RESOLVE BEFORE DEVELOPMENT STARTS

The following items must be confirmed and received from Trackpilots team (team@trackpilots.com) before Soniq backend development begins. Without these, webhook integration and API integration cannot be built or tested.

2.3 Tech Stack Decision (Recommended for Phase 1)

Layer	Technology	Rationale
Backend Runtime	Node.js (v20 LTS) with TypeScript	Fast, event-driven — ideal for webhook processing; large ecosystem
Web Framework	Express.js	Lightweight, flexible — easy webhook endpoint setup with raw body middleware
Job Queue	BullMQ + Redis	Reliable webhook event queue — handles retry, priority, concurrency

Layer	Technology	Rationale
Primary Database	PostgreSQL 15	Robust relational DB — complex queries for reports and aggregations
ORM	Prisma	Type-safe DB access, auto-migration, Postgres support
Screenshot Storage	AWS S3 + CloudFront CDN	Industry standard for image storage; signed URLs for secure access
Image Processing	Sharp (Node.js)	Fast server-side image blurring for privacy mode screenshots
Real-time Updates	Socket.io / SSE	Push live dashboard updates (work mode changes) to connected clients
Frontend Framework	Next.js 14 (React)	SSR for performance; file-based routing; two separate Next.js apps for portals
Styling	Tailwind CSS + Shadcn UI	Rapid, consistent UI development; Poppins font via Google Fonts
Charts/Visualizations	Recharts / Apex Charts	Rainbow semicircle, bar charts, timeline visualization
Auth	JWT + NextAuth.js	Stateless auth with refresh tokens; separate auth for each portal
Email	Resend API	Modern transactional email with React Email templates
Environment Config	Doppler / dotenv-vault	Secure secret management across environments
Testing	Jest + Supertest (backend) / Playwright (E2E)	Full test coverage before May launch
Deployment	Docker + Docker Compose	Containerized deployment; easy to replicate staging → production

3. Super Admin Portal (admin.soniq.io)

SEPARATE PORTAL — INTERNAL USE ONLY

The Super Admin Portal is a completely separate Next.js application from the Client Portal. It has its own authentication, its own API routes, its own database queries, and its own UI. Client Admins CANNOT access this portal. Super Admins CANNOT access the Client Portal except through impersonation (audit-only, logged). URL: admin.soniq.io

3.1 Super Admin Authentication

- Separate login page at admin.soniq.io/login
- Email + Password login — no SSO in Phase 1
- Super Admin accounts are created manually by the Soniq engineering team — no self-registration
- JWT tokens with 8-hour expiry; refresh token with 7-day sliding window
- All Super Admin actions are written to the audit log with timestamp, actor email, action taken, target org/entity

3.2 Super Admin Dashboard

Platform-wide health and business metrics at a glance.

Summary Metric Cards

Card	Metric	Source Table
Total Client Organizations	COUNT of all orgs	organizations
Active Clients	COUNT where status = active	organizations
Suspended Clients	COUNT where status = suspended	organizations
Trial Clients	COUNT where status = trial	organizations
Total Employees Monitored	SUM of all active employees across all orgs	employees
Employees Online Right Now	COUNT with webhook event in last 5 min	activity_events (Redis cache)
Webhooks Received Today	COUNT of all events received today	activity_events
Webhook Error Rate Today	Failed events / total events × 100%	webhook_logs
Platform MRR	SUM of active subscription monthly amounts	subscriptions
Clients Due Renewal (30 days)	COUNT expiring within 30 days	subscriptions

Platform Activity Charts

- Webhook Volume Chart: Daily bar chart — events received per day for last 30 days (broken by event type: activity/app/screenshot)
- Client Growth Chart: Line chart — cumulative client org count over time
- Error Rate Chart: Line chart — webhook processing error % over last 7 days
- Top Active Clients Today: Table — Org Name | Active Employees | Screenshots Today | Webhooks Today

3.3 Client Organization Management

Organization List Table

Column	Description	Source
Organization Name	Company name + avatar initials	organizations.name
Status Badge	active (green) trial (blue) suspended (red) inactive (gray)	organizations.status
Plan	Basic Professional Enterprise	subscriptions.plan_name
Total Employees	Active employee count	COUNT(employees)
Onboarded On	First activation date	organizations.created_at
Last Webhook	Most recent webhook received from this org	MAX(activity_events.received_at)
MRR	Monthly revenue from this client	subscriptions.monthly_amount
Renewal Date	When their plan expires	subscriptions.expires_at
Actions	View Edit Suspend Reactivate Delete	—

Organization Detail Page — Six Tabs

Tab	Content
Overview	Company profile: name, industry, website, country, phone, timezone, Trackpilots org ID (masked), API key (masked with reveal button), webhook secret (masked), plan details, onboard date, contract notes
Employees	Full employee list for this org: same columns as client-facing view. Super Admin can view any employee's data, navigate to their detail page, force-remove an employee
Teams	All teams in the org: Team Name Member Count Created Date Actions (Edit/Delete)
Settings	Super Admin override panel: enable/disable specific feature modules per org (stealth monitoring, advanced reports, screenshot blur), data retention policy override, storage quota
Billing	Plan type, billing cycle, amount, payment history (date amount status), next renewal date, upgrade/downgrade plan selector, manual invoice generation

Tab	Content
Audit Log	Every action taken on this org: timestamp actor action before value after value. Includes: org created, plan changed, employee added/removed, settings changed, suspension events

3.4 Super Admin — Employee Global View

Super Admins can search and view any employee across any organization for support, compliance, and audit purposes.

Column	Description
Employee Name + Email	Full name and email with avatar
Organization	Which client org this employee belongs to (clickable)
Team	Assigned team name
Role	Their access role within their org
Status	active inactive
OS	windows macos linux — from last webhook event
EXE Version	Trackpilots agent version installed on their machine
Last Seen	Timestamp of last webhook event received
Current Mode	Work Mode Privacy Mode Offline App Not Installed (live from Redis cache)
Actions	View Activity Timeline View Screenshots Impersonate (logged) Deactivate

3.5 Super Admin — Subscription & Billing Management

Plan Configuration

Plan	Features	Limits
Basic	Dashboard, Teams, Screenshots, Attendance, Basic Reports (Productivity Insights + Attendance), Settings	Max 50 employees, 30-day data retention, 10GB screenshot storage, no stealth monitoring

Plan	Features	Limits
Professional	Everything in Basic + All 5 Reports, Advanced Settings, Effort Utilization	Max 200 employees, 90-day retention, 50GB storage, stealth monitoring available
Enterprise	Everything in Professional + Custom data retention, dedicated support, white-label options	Unlimited employees, custom retention, unlimited storage, custom integrations

Billing Dashboard Metrics

- MRR (Monthly Recurring Revenue) — total across all active subscriptions
- ARR (Annual Recurring Revenue) — projected
- Churn Rate — percentage of clients lost in the last 30 days
- Plan Distribution — pie chart: how many clients on each plan
- Upcoming Renewals — list of orgs renewing in next 7/15/30 days with reminder send button

3.6 Super Admin — Platform Settings & Health

Setting	Description
Global Productivity Classifications	Define default app/website category productivity status (productive/neutral/unproductive) applied to all orgs. Orgs can override per their settings.
Feature Flags per Org	Toggle specific features on/off per client org (e.g. enable stealth monitoring for Enterprise org only)
Webhook Monitor	Real-time table: last 100 webhook events across all orgs — event type, org, employee, status (success/fail), latency in ms
Queue Monitor	BullMQ dashboard: webhook processing queue depth, active jobs, failed jobs, retry count
Storage Monitor	S3 usage per org — screenshots stored, total size, % of quota used
Error Log	All backend errors: timestamp, error type, stack trace, affected org/employee, resolution status
System Health	Server CPU %, RAM usage, DB connection pool, Redis memory, API response time P50/P95/P99
Notification Templates	Edit email templates for: client invite, plan renewal reminder, suspension notice, welcome email

4. Client Portal Overview (app.soniq.io)

SEPARATE PORTAL — FOR CLIENT ORGANIZATIONS

The Client Portal is a separate Next.js application from the Super Admin Portal. Every client organization's Admin and Managers log in here. Data is fully scoped to the logged-in user's organization — an Admin from Org A can never see Org B's data. URL: app.soniq.io

4.1 Client Portal Authentication

- Login URL: app.soniq.io/login
- Email + Password — Client Admin sets up account during Soniq onboarding
- JWT access token (4-hour expiry) + refresh token (30-day sliding window)
- All API requests include org_id scoping — backend enforces data isolation at query level
- Forgot Password: Email OTP flow via Resend/SendGrid
- First-time login: Prompt to set password if invited by Super Admin

4.2 Client Portal Navigation Structure

Module	Sub-pages / Tabs	Available To
Dashboard	Main dashboard with all widgets	Client Admin, Client Manager
Teams	Team List → Team Detail → Employee Detail	Client Admin, Client Manager (own teams only)
Screenshots	Screenshot grid with filters	Client Admin, Client Manager
Attendance	Attendance List Timeline View	Client Admin, Client Manager
Reports	Hub → Productivity Insights Effort Utilization Attendance App/Web Usage Timesheet	Client Admin only (Manager: read-only if role permits)
Settings	Manage Team Manage Employee Work Day Expected Work Hours Screenshot Idle Alert Stealth Monitoring Default Settings Access Control	Client Admin only

4.3 Client Onboarding Flow on Soniq

- Step 1: Super Admin creates a new organization in admin.soniq.io → enters org name, plan, contact email
- Step 2: Soniq sends a welcome email to the Client Admin email with a one-time setup link
- Step 3: Client Admin sets their password and logs into app.soniq.io
- Step 4: Client Admin goes to Settings → Manage Employee → adds employees (triggers Trackpilots invite via Trackpilots API)
- Step 5: Employees receive Trackpilots invite email, create their Trackpilots account, download and install the EXE
- Step 6: EXE starts sending webhook events to Soniq — data begins appearing on dashboard within minutes
- Step 7: Client Admin configures teams, work hours, screenshot settings as needed

5. Client Portal — Module 1: Dashboard

The Dashboard is the first screen after login for every Client Admin and Client Manager. It provides a real-time, comprehensive snapshot of the organization's or team's productivity. It is the most complex and most important module in Soniq.

TRACKPILOTS REST APIS USED

GET /v1/dashboard/employee-recent-activity | POST /v1/dashboard/fetch-work-summary | POST /v1/dashboard/fetch-top-productive-performer | POST /v1/dashboard/fetch-recent-screenshot | POST /v1/dashboard/fetch-work-time-analysis | POST /v1/dashboard/fetch-productivity-analysis | POST /v1/dashboard/fetch-apps-and-websites | POST /v1/dashboard/fetch-weekly-summary | POST /v1/dashboard/fetch-work-time-efficiency

5.1 Dashboard — Top Bar Filters

Filter	UI Element	Options	Effect
Business Hours Mode	Toggle switch	Business Hrs: 8hr/Day (ON/OFF)	When ON: productivity measured against configured expected hours. When OFF: uses actual tracked time
Date Range	Date picker with presets	Today This Week Custom Range	All dashboard widgets refresh for the selected period
Team Filter	Dropdown	All Teams [Each team name]	Filters all widgets to show data for selected team only. Client Manager sees only their assigned team(s)

5.2 Summary Stat Cards — 4 Cards (Top Row)

Four horizontally spaced metric cards. Each card has a large primary number, a label, and a percentage change indicator vs the previous equivalent period.

Card #	Label	Primary Value	Sub Stat / Trend	DB Source
Card 1	Total Employees	Total employee count in the selected team/org filter	— (static count for filter scope)	COUNT(employees WHERE org_id AND team_id)
Card 2	Present Employees	Count of employees with at least one activity_event today	% of total present (e.g. 94/128 = 73.4%)	COUNT DISTINCT employees in daily_summaries WHERE is_present=true AND date=today
Card 3	Absent Employees	Count of employees absent on a working day	Delta vs yesterday (e.g. +5 more absent than yesterday)	Total - Present (on configured working days only)
Card 4	Avg Productivity / Day	Average productive hours per day across all present employees for date range (e.g. 6h 42m)	% change vs previous period	AVG(daily_summaries.productive_seconds) for date range

- Trend indicator: Green arrow up = positive, Red arrow down = negative, Gray = no change
- Cards are non-clickable display metrics in Phase 1 — no drill-down navigation
- Cards refresh every 60 seconds via polling (not real-time SSE — only live scroller uses SSE)

5.3 Work Summary — Rainbow Semicircle Chart

A visually distinctive semicircle donut chart positioned at the top right of the dashboard. Five colored arc segments represent different time states. The center displays the total working hours prominently.

Segment	Color (Hex)	Label	Source Field
Productive Time	#16A34A (Green)	Productive Hrs	SUM(activity_events.duration_seconds WHERE productivity_status='productive')
Neutral Time	#2563EB (Blue)	Neutral Hrs	SUM(activity_events.duration_seconds WHERE productivity_status='neutral')

Segment	Color (Hex)	Label	Source Field
Idle Time	#CA8A04 (Amber)	Idle Hrs	daily_summaries.idle_seconds (computed gap between events)
Unproductive Time	#DC2626 (Red)	Unproductive Hrs	SUM(activity_events.duration_seconds WHERE productivity_status='unproductive')
Total Working Hours	#0F2044 (Navy)	Total Working Hrs	daily_summaries.total_work_seconds (first event to last event)

- Center label: Total Working Hours (e.g. '8h 00m') in large bold Poppins font
- Each arc thickness is proportional to its time share of the total working hours
- Legend below chart: colored dot + label + formatted duration (e.g. Productive 5h 13m)
- Two quick stats beside chart: 'Available Hrs: 16' and 'Shared Hrs: 32 / Last Hours: 9h 55m'
- Chart uses SVG — rendered client-side with Recharts RadialBarChart or ApexCharts

5.4 Live Status Scroller (Real-Time Ticker)

A live auto-scrolling notification feed positioned at the bottom right. Shows real-time employee mode changes as they occur, driven by incoming webhook events pushed via Server-Sent Events (SSE) from Soniq's backend.

Status	Display	Webhook Source	Visual Indicator
Work Mode	Employee is actively working — desktop app tracking	activity.workMode = true	Green dot + 'Work Mode' badge
Privacy Mode	Employee paused tracking manually	activity.workMode = false (app still running)	Orange dot + 'Privacy Mode' badge
Offline	No webhook event received for > configured idle threshold	Computed server-side: no events in last 5 min	Gray dot + 'Offline' badge + last seen time
App Not Installed	Employee exists but never sent a webhook event	trackpilots_user_id mapped but no events ever	Red dot + 'App Not Installed' badge

- Each entry: Employee avatar (initials if no profile pic) | Employee name | Status badge | Timestamp (HH:MM)
- New entries scroll up from bottom — oldest entries fade out at top
- Clicking an entry navigates to that employee's detail page
- SSE endpoint: GET /api/dashboard/live-status — streams events to connected clients

- Backend listens for incoming activity_tracking webhook events and broadcasts to all connected SSE clients for that org

5.5 Top Productive Panel

Shows the top 5 highest-performing employees, websites, or apps by productive time. Three toggle buttons: [Employee] [Website] [App]. Default active tab: Employee.

Tab	Row Contents	Bar	Sort Order	DB Query
Employee	Avatar Full Name Horizontal bar Productive Hrs:Min:Sec	Green bar proportional to max performer	DESC by productive_seconds	daily_summaries grouped by employee_id for date range
Website	Favicon Domain Name Horizontal bar Time Spent	Green bar	DESC by total duration_seconds	activity_events WHERE app_type='website' AND productivity_status='productive'
App	App Icon App Name Horizontal bar Time Spent	Green bar	DESC by total duration_seconds	activity_events WHERE app_type='application' AND productivity_status='productive'

- Top 5 entries only — 'View All' link at bottom navigates to Productivity Insights report
- Bar width: top performer = 100% width; others proportional (e.g. if #2 has 70% of #1's time → 70% bar width)
- Time format: 7h 14m 33s — hours, minutes, seconds all shown

5.6 Top Unproductive Panel

Same layout as Top Productive panel but for unproductive time. Default active tab: App (because apps are the most common source of unproductive time).

Tab	Row Contents	Bar Color	Default?
Employee	Avatar Name Bar Unproductive Hrs	Red (#DC2626)	No
Website	Favicon Domain Bar Time Spent	Red	No

Tab	Row Contents	Bar Color	Default?
App	App Icon App Name Bar Time Spent	Red	YES — default active tab

5.7 Today's Activity Table (Live)

Real-time table updated every 30 seconds (or via SSE). Shows all employees' current day status.

Column	Description	Source
Employee	Profile avatar + Full name — clicking navigates to employee detail	employees table
Login Time	First activity_event of the day — displayed as HH:MM	MIN(activity_events.start_time) for today
Active Hrs	Total hours employee has been in work mode today	SUM(duration_seconds WHERE workMode=true) from activity_events
Progress Bar	Visual bar: active hours / expected work hours × 100%. Color: Green >80%, Orange 50-80%, Red <50%	Computed vs expected_work_hours setting
Score %	Productivity score: productive_seconds / (productive + unproductive + neutral) × 100	daily_summaries.productivity_score

- Toggle: [Live] [Weekly] — Live shows today's real-time rows; Weekly shows 7-day summary condensed
- Employees sorted by productivity score descending by default
- Employees absent today shown at bottom with 'Absent' badge and no data

5.8 Work Hour Insight Bar Chart

Daily grouped bar chart showing Productive vs Unproductive hours for each day in the selected date range. Positioned at the bottom right of the dashboard.

Element	Description
X-Axis	Dates in selected range (Jan 01, Jan 02, Jan 03 ...)
Y-Axis	Hours (0 to max of productive + unproductive for the range)

Element	Description
Productive Bars	Navy/Blue filled bars — productive hours per day
Unproductive Bars	Light blue / outlined bars — unproductive hours per day
Tooltip on Hover	Date Productive: X hr Y min Unproductive: X hr Y min
Legend	● Productive ○ Unproductive (top right of chart)

5.9 Dashboard — Missing Elements from Original BRD (Now Added)

ADDITIONAL DASHBOARD FEATURES

The following were identified as missing from the previous BRD version and are now included:

- Work Mode Status Summary Bar (below top cards): Total Members | Working Mode count | Privacy Mode count | Offline count | App Not Installed count — as clickable filter chips that filter the Today's Activity table
- Recent Screenshots Strip: A horizontal strip of the 5 most recent screenshots across the org — thumbnail + employee name + time. Clicking opens the full Screenshots module
- Team Productivity Comparison (if multiple teams): Small bar chart comparing avg productivity score per team for the day — helps admin identify underperforming teams at a glance
- Notification Bell Icon (top right): Shows count of unread alerts (idle time exceeded, app-not-installed employees, employees offline > 30 min)

6. Client Portal — Module 2: Teams

The Teams module gives Client Admins and Managers visibility into team-level productivity data. It has three levels: Team List → Team Detail (with Calendar) → Individual Employee Detail.

TRACKPILOTS APIS USED

POST /v1/my-team/fetch-my-team | GET /v1/my-team/fetch-time-logs | GET /v1/my-team/fetch-work-summary | GET /v1/my-team/fetch-screenshot | GET /v1/my-team/fetch-apps-and-websites-usage

6.1 Team List View

Column	Description	Source
Team Name	Team name with color-coded avatar (initials)	teams.name
Total Members	Count of employees assigned to this team	COUNT(employees WHERE team_id)
Active Now	Count with activity event in last 5 minutes (live)	Redis cache: live status by team
Working Mode	Count currently in work mode	Live from activity tracking events
Privacy Mode	Count in privacy mode	Live from activity tracking events
Offline	Count with no event in last 5 min	Computed
Avg Productivity Today	Team average productivity score for today	AVG(daily_summaries.productivity_score) WHERE date=today AND team_id
Actions	View Team Detail	Navigation

- Status filter tabs: [All] [Active Members] [Working Mode] [Privacy Mode] [Offline]
- Search: Filter team list by team name
- If Client Manager is logged in: shows only their assigned teams

6.2 Team Detail — Summary Cards

Card	Metric	Source
Total Employees	Member count in this specific team	COUNT(employees WHERE team_id)
Avg Productivity / Day	Average productive hours per day for date range	AVG(daily_summaries.productive_seconds) for team + date range
Available Hours	Expected work hours × present employee count	expected_work_hours × COUNT(is_present=true)
Total Absent Hours	Absent employee count × expected work hours	absent_count × expected_work_hours

6.3 Team Detail — Work Hour Calendar View

A monthly calendar heat map showing daily work hours for the team. Toggle between months with previous/next arrows.

Element	Description
Calendar Grid	Monthly grid — each day cell shows the team's average productive hours for that day
Color Coding	Dark Green = excellent (> 80% target) Light Green = good (60-80%) Yellow = below target (40-60%) Red = poor or mostly absent (<40%) Gray = non-working day / weekend / holiday
Cell Content	Date number + avg hours (e.g. '6.4h')
Click Behavior	Click any day → filters the employee table below to show data for that specific date
Navigation	◀ Previous Month ▶ Next Month
Legend	Color scale bar at bottom of calendar

6.4 Team Detail — Employee List Table

Column	Description	Source
Employee Name	Avatar + Full name — clickable → employee detail page	employees.name + profile_pic_url
Status	active (green badge) Unregistered (red badge = invited but EXE not installed)	employees.status + presence of webhook events
Productive Hours	Total productive hours for date range + target indicator badge (e.g. 65% of target)	daily_summaries.productive_seconds vs expected_productive_work_hours
Check In	First activity time of the selected day (HH:MM)	MIN(activity_events.start_time)
Check Out	Last activity time of the selected day (HH:MM)	MAX(activity_events.end_time)
Unproductive	Total unproductive time formatted as h:mm:ss	daily_summaries.unproductive_seconds
Away Time	Idle/away time computed from event gaps	daily_summaries.idle_seconds

Column	Description	Source
Idle Time	Explicit idle detection from Trackpilots	SUM(duration_seconds WHERE productivity_status='idle')
Most Used App/Site	Top app or domain by time spent for the period	activity_events grouped by app_name, top 1
Team	Team name (relevant in org-wide views)	teams.name
Actions	View Details Remove from Team	Navigation + team assignment

- View toggle: [Detailed View] [Calendar View] — Detailed = table, Calendar = heat map
- Date picker: Select single date or date range
- Export button: Download as CSV / Excel — all rows, not paginated
- Pagination: 25 rows per page

6.5 Individual Employee Detail Page

Accessible from anywhere in Soniq where an employee name is clickable. Shows the complete productivity breakdown for one employee.

Left Panel — Employee Profile Card

Field	Description
Avatar	Profile picture from Trackpilots or initials fallback
Full Name	Employee full name
Employee Code	Internal ID or Trackpilots userId (masked partially)
Current Status Badge	Work Mode (green) Privacy Mode (orange) Offline (gray) App Not Installed (red) — live
Recent App/Website	Most recently active app or website with its productivity badge
Daily Average	Average productive hours per day for selected range
Check-In Time	Today's first activity timestamp
Check-Out Time	Today's last activity timestamp (or 'Still Active' if online)
Productivity Score	Circular gauge 0-100 with label: Excellent (≥ 85) Good (70-84) Average (50-69) Needs Improvement (< 50)

Right Panel — Four Tabs

Tab	Content Description
Overview	Productivity Summary (4 cards: Productive Time Unproductive Time Away Time Idle Time). Attendance Summary (Working Days Absent Days EXE Issue Days). Time Tracker Chart (hourly bar chart). Productive Usage table (Apps + Websites sub-tabs). Unproductive Usage table. Categories pie chart.
Meetings	Placeholder — Out of scope Phase 1
Attendance	Daily attendance record table: Date Check-In Check-Out Total Hours Productive Unproductive Idle Status
Regularization	Placeholder — Out of scope Phase 1

Overview Tab — Time Tracker Chart

- Horizontal bar timeline chart for the selected date
- X-axis: Time of day (00:00 to 23:59, showing working hours range)
- Y-axis: Single bar for the employee (or per-app bars)
- Segments: Productive (green) | Unproductive (red) | Neutral (blue) | Idle (amber) | Away (gray)
- Target line: Vertical dotted line at expected check-out time
- Tooltip: Hover over segment → App name | Category | Duration | Start time - End time

Overview Tab — Productive / Unproductive Usage Tables

Column	Applications Sub-tab	Websites Sub-tab
Name / Domain	App name with icon	Website domain with favicon
Usage Time	Time in hh:mm:ss	Time in hh:mm:ss
% of Total	% share of active time	% share of active time
Category	e.g. Graphics & Multimedia	e.g. Computers and Technology
Status Badge	Productive Neutral Unproductive	Productive Neutral Unproductive

7. Client Portal — Module 3: Screenshots

The Screenshots module provides visual proof of employee work. All images are received via the `desktop.screenshot_tracking.captured` webhook, stored in Soniq's own S3 bucket (not Trackpilots storage), and served via CDN from Soniq's URLs.

SCREENSHOT STORAGE FLOW

Trackpilots EXE captures screenshot → sends webhook with base64 imageBuffer (up to 50MB) → Soniq backend receives, HMAC-validates, decodes imageBuffer → optionally applies Gaussian blur via Sharp library if blur is enabled for the employee → uploads JPEG to AWS S3 at path: `soniq-screenshots/{org_id}/{employee_id}/{YYYY-MM-DD}/{screenshotid}.jpg` → stores metadata (not the binary) in screenshots table with S3 URL → serves via CloudFront CDN with signed URL expiry

7.1 Screenshot Filters

Filter	UI Element	Options	Effect
View By	Toggle	[User] [Team]	Switch between individual and team-wide view
Team/Employee	Dropdown	All Teams [Team Name] [Employee Name]	Filter to specific team or person
Date	Date picker	Single date or date range	Show screenshots for selected date(s) only
Productivity Status	Toggle buttons (multi-select)	[Productive] [Neutral] [Unproductive] [Idle]	Filter grid by app's productivity status at capture time

7.2 Screenshot Grid Layout

Element	Description	Source
Thumbnail Image	Scaled JPEG preview rendered from Soniq CDN URL (not S3 direct)	<code>screenshots.image_url</code> → CloudFront signed URL
Employee Name	Name and avatar below thumbnail	<code>employees.name</code>
Capture Time	HH:MM:SS format	<code>screenshots.captured_at</code>

Element	Description	Source
Active App/Site	App name or domain at capture time	screenshots.app_name or screenshots.app_domain
Productivity Badge	Color-coded pill: Productive (green) Neutral (blue) Unproductive (red) Idle (amber)	screenshots.productivity_status
OS Icon	Windows or macOS icon in corner of thumbnail	screenshots.operating_system
Idle Overlay	If is_idle=true: amber warning icon overlay on top of thumbnail + 'Idle' label	screenshots.is_idle
Blur Indicator	If screenshot was blurred: blur icon overlay indicating privacy mode image	screenshots.is_blurred (stored in DB)

- Grid layout: 4 thumbnails per row on desktop, 2 on tablet
- Clicking any thumbnail → opens full-screen modal
- Pagination: 24 per page with infinite scroll option

7.3 Screenshot Full-Screen Modal

Element	Description
Full Image	Full resolution image served from CDN with signed URL (expires in 1 hour)
Employee Info	Avatar + Name + Team
Capture Details	Date Time App Name Full URL Category Productivity Status
System Info	OS: Windows/macOS Work Type: Office/Remote/Hybrid
Idle Status	'Employee was idle at time of capture' warning if is_idle=true
Navigation	◀ Previous Screenshot ▶ Next Screenshot within the current filtered set
Close	X button or Escape key

7.4 Screenshots Database Schema

DB Column	Type	From Webhook Field	UI Usage
id	UUID PK	—	Internal reference
org_id	UUID FK	webhook.data[.].organisation.organisationId → mapped	Data isolation
team_id	UUID FK	webhook.data[.].team.teamId → mapped	Team filter
employee_id	UUID FK	webhook.data[.].user.userId → mapped	Employee filter
trackpilots_screenshot_id	VARCHAR	screenshot.screenshotId	Deduplication
image_url	VARCHAR	imageBuffer → decoded → uploaded to S3 → S3 URL stored	Thumbnail + full image
is_blurred	BOOLEAN	Computed: check employee screenshot settings at time of receipt	Blur indicator badge
app_name	VARCHAR	screenshot.app.name	Active app display
app_type	VARCHAR	screenshot.app.type (application/website)	Filter
app_category	VARCHAR	screenshot.app.category	Category display
app_domain	VARCHAR	screenshot.app.domain	URL display
app_full_url	TEXT	screenshot.app.fullUrl	Full URL in modal
productivity_status	ENUM	screenshot.app.productivityStatus	Productivity badge + filter
is_idle	BOOLEAN	screenshot.isIdle	Idle overlay
captured_at	TIMESTAMP	screenshot.time.capturedAt	Time display + date filter
operating_system	VARCHAR	screenshot.operatingSystem	OS icon
work_type	VARCHAR	screenshot.workType (hybrid/remote/office)	Work type in modal

8. Client Portal — Module 4: Attendance

Attendance data is fully derived from Trackpilots webhook events. There is no manual check-in mechanism. The first activity_tracking event of the day = check-in. The last activity_tracking event = check-out. All attendance metrics are computed by Soniq's daily aggregation job.

ATTENDANCE COMPUTATION ENGINE

Soniq runs a scheduled job (cron: every 5 minutes for live data, midnight for daily finalization) that:

- 1) Groups all activity_events by employee_id + date.
- 2) Computes first_checkin, last_checkout, total_work_seconds.
- 3) Aggregates productive/unproductive/neutral/idle seconds.
- 4) Sets is_present = true if any events exist for that employee on that working day.
- 5) Sets is_present = false (absent) if no events on a configured working day.
- 6) Updates daily_summaries table.

8.1 Attendance — List View (Default)

Column	Description	Source
Employee Name	Avatar + Full name — clickable to employee detail	employees.name
Status Badge	Present (green) Absent (red) EXE Issue (orange — employee exists but zero events ever) Weekend (gray) Holiday (gray)	Computed from daily_summaries.is_present + work_day_settings
Shift	Configured shift name and hours (e.g. Night Shift 10:00 PM – 07:00 AM) — if configured in Expected Work Hours settings	expected_work_hours.expected_in_time + expected_work_hours.expected_work_hours
Timeline Bar	Horizontal bar spanning configured working hours (e.g. 7 AM to 9 PM). Each segment colored by activity: Productive=Green Unproductive=Red Neutral=Blue Idle=Amber Away=Gray Break=Orange Shift Start marker Shift End marker	Plotted from activity_events for the day on the time axis
Check-In	First activity timestamp (HH:MM)	daily_summaries.first_checkin
Check-Out	Last activity timestamp (HH:MM) or 'Active' if still working	daily_summaries.last_checkout
Total Hours	Duration: last_checkout - first_checkin	Computed

Column	Description	Source
Productive	Productive seconds formatted h:mm:ss	daily_summaries.productive_seconds
Unproductive	Unproductive seconds formatted h:mm:ss	daily_summaries.unproductive_seconds
Idle	Idle time formatted h:mm:ss	daily_summaries.idle_seconds

- Tab filters: [All (N)] [Present (N)] [Absent (N)] — N shows count
- Date navigation: ◀ Previous Day | Date picker | ▶ Next Day
- Search: Filter by employee name
- Color legend: Productive | Unproductive | Regularized | Unclassified | Idle | Away | Break Time | Worked During Break | Shift Start | Shift End
- Export: CSV / Excel download of all rows for selected date

8.2 Attendance — Timeline View

A detailed hour-by-hour 24-hour timeline for all employees on the selected date. Accessed via a tab toggle [List View] [Timeline View].

Element	Description
Time Axis (X)	Full 24-hour ruler at top: 12am 1am 2am ... 11pm (horizontally scrollable)
Employee Rows (Y)	One row per employee. Employee name + check-in time on left side
Activity Segments	Each contiguous block of activity plotted as a colored segment on the employee's row: Productive (green) Unproductive (red) Neutral (blue) Idle (amber) Away (gray)
Shift Markers	Vertical dotted line at configured shift start and shift end times
Break Time	Orange segment for detected break periods (no events for > 15 minutes within working hours)
Worked During Break	Teal segment if employee was active during a configured break window
Tooltip on Segment Hover	App Name Category Duration Start HH:MM → End HH:MM Productivity Status
Filters	User Team toggle + Team/Employee dropdown + Productivity status multi-select buttons
Navigation	◀ Previous Day Date Picker ▶ Next Day
Current Time Line	Vertical red line showing current time (live, moves in real-time)

8.3 Attendance — Key Computed DB Fields

Field	Table	Computation Logic
is_present	daily_summaries	true if COUNT(activity_events) > 0 for employee on date AND date is a configured working day
first_checkin	daily_summaries	MIN(activity_events.start_time) for employee on date
last_checkout	daily_summaries	MAX(activity_events.end_time) for employee on date
total_work_seconds	daily_summaries	EXTRACT(EPOCH FROM last_checkout - first_checkin)
idle_seconds	daily_summaries	total_work_seconds - productive_seconds - unproductive_seconds - neutral_seconds
productivity_score	daily_summaries	(productive_seconds / (productive_seconds + unproductive_seconds + neutral_seconds)) × 100 — NULL if all zero
is_late	daily_summaries	first_checkin > expected_in_time for the employee + 10 minute grace period
overtime_seconds	daily_summaries	MAX(0, total_work_seconds - expected_work_hours_in_seconds)
is_exe_issue	employees	true if employee exists and zero webhook events ever received for them

9. Client Portal — Module 5: Reports

The Reports module provides five structured analytical reports. All reports are served from Soniq's processed database (daily_summaries + activity_events) — not from Trackpilots APIs directly for performance reasons. All reports support Employee / Team / Department grouping, date range filtering, Day Wise / Summary toggle, and CSV/Excel export.

9.1 Reports Hub — Landing Page

Card grid showing all available reports. Each card: icon + report name + description + 'View Report' arrow button.

Report	Description on Card
Productivity Insights	View detailed insights into productive, unproductive, and idle work time per employee

Report	Description on Card
Effort Utilization	Understand how employee effort maps to actual productive output – identifies high vs low utilizers
Attendance Report	Track check-in, check-out, working days, absent days with full productivity context
App / Website Usage	Detailed breakdown of productive and unproductive app and website usage per employee
Employee Timesheet Report	Comprehensive timesheet: productive hours, idle time, check-in/out, most used apps
Site Search Report (Optional)	Analytics on frequently searched terms to understand user needs – implement if time permits

9.2 Common Features — All Reports

Feature	Description
Scope Toggle	[Employee] [Team] [Department] – changes the grouping/filtering of data
Date Range	Date picker: From → To with calendar. Presets: Today This Week This Month Last Month Custom
View Toggle	[Day Wise] [Summary] – Day Wise = one row per employee per day; Summary = one aggregate row per employee for full range
Search	Text input to filter table rows by employee name
Sort	Click any column header to sort ascending / descending
Pagination	25 rows per page with page navigation (< 1 2 3 ... >)
Export	Button: [↑ Export] – Downloads all rows (not just current page) as CSV or Excel (.xlsx)
Column Visibility	Optional: Hide/show columns via column selector

9.3 Report 1 — Productivity Insights

Table Columns — Day Wise

Column	Description	Source
Date	Date of the record	daily_summaries.summary_date
Name	Employee full name	employees.name
Designation	Job title (if configured in employee profile)	employees.designation
Team	Team name	teams.name
Business Hours	Configured expected work hours for that employee on that day	expected_work_hours.expected_work_hours_per_day
Productivity Target %	Configured productivity target (e.g. 82%)	org or employee-level setting
Actual Productivity %	$\text{productive_seconds} / (\text{productive} + \text{unproductive} + \text{neutral}) \times 100$	daily_summaries.productivity_score
Productive Time	Formatted as 00h 00m 00s (% of total active time)	daily_summaries.productive_seconds
Unproductive Time	Formatted as 00h 00m 00s (% of total)	daily_summaries.unproductive_seconds
Idle Time	Formatted as 00h 00m 00s (% of total)	daily_summaries.idle_seconds

9.4 Report 2 — Effort Utilization

Table Columns

Column	Description	Source
Date	Date of record	daily_summaries.summary_date
Name	Employee name	employees.name
Designation	Job title	employees.designation
Team	Team name	teams.name

Column	Description	Source
Most Used App/Site	Top app or domain by time for that day	activity_events top by duration_seconds per employee per day
Usage Time	Time spent on the most used app/site	SUM(activity_events.duration_seconds) for top app
Effort Utilization	High Effort Utilization Normal Effort Utilization Low Effort Utilization — computed: High = productive% > 80%, Normal = 50-80%, Low = <50%	Computed from daily_summaries.productivity_score
Productive Status	Excellent (≥85%) Good (70-84%) Average (50-69%) Needs Improvement (<50%)	daily_summaries.productivity_score banded
Total Productive Time	Formatted 00h 00m 00s % of day	daily_summaries.productive_seconds
Total Unproductive Time	Formatted 00h 00m 00s % of day	daily_summaries.unproductive_seconds
Total Idle Time	Formatted 00h 00m 00s % of day	daily_summaries.idle_seconds

9.5 Report 3 — Attendance Report

Column	Description	Source
Date	Date of record	daily_summaries.summary_date
Name	Employee name	employees.name
Designation	Job title	employees.designation
Team	Team name	teams.name
Business Day	1 = configured working day 0 = weekend or holiday	work_day_settings + holidays
Present Day	1 = employee had activity 0 = no activity on working day	daily_summaries.is_present
Absent Day	1 = working day and not present 0 otherwise	Computed
Check-In Time	First activity timestamp for the day	daily_summaries.first_checkin
Check-Out Time	Last activity timestamp for the day	daily_summaries.last_checkout

Column	Description	Source
Business Hours	Configured expected hours	expected_work_hours settings
Productivity Target %	Configured target	settings
Actual Productivity %	Computed score	daily_summaries.productivity_score
Productive Time	h:mm:ss formatted	daily_summaries.productive_seconds
Unproductive Time	h:mm:ss formatted	daily_summaries.unproductive_seconds
Idle Time	h:mm:ss formatted	daily_summaries.idle_seconds

9.6 Report 4 — App / Website Usage

Column	Description	Source
Date	Date of record	activity_events.start_time::date
Name	Employee name	employees.name
Team	Team name	teams.name
Department	Department if configured	employees.department
Application / Website	App name or domain	activity_events.app_name or app_domain
Type	application website	activity_events.app_type
Category	e.g. Computers and Technology Social Media Entertainment Graphics & Multimedia	activity_events.app_category
Usage Time	Total time on this app/site for that employee on that day (h:mm:ss)	SUM(activity_events.duration_seconds) grouped by employee + app + date
Productivity Status	productive neutral unproductive — color-coded badge	activity_events.productivity_status

9.7 Report 5 — Employee Timesheet Report

Column	Description	Source
Date	Date of record	daily_summaries.summary_date
Name	Employee name	employees.name
Team	Team name	teams.name
Department	Department if configured	employees.department
Idle Time	Idle hours formatted h:mm:ss	daily_summaries.idle_seconds
Productive Time	Productive hours formatted h:mm:ss	daily_summaries.productive_seconds
Unproductive Time	Unproductive hours formatted h:mm:ss	daily_summaries.unproductive_seconds
Check-In Time	First login time	daily_summaries.first_checkin
Check-Out Time	Last logout time	daily_summaries.last_checkout
Most Used App/Site	Top app or domain by time for that day	activity_events top 1 by duration per employee per day
Target %	Configured productivity target %	settings
Productive %	Actual productivity score	daily_summaries.productivity_score

10. Client Portal — Module 6: Settings

Settings allow Client Admins to configure organizational policies, employee assignments, and access control. Settings are stored in Soniq's database. Where applicable, changes are pushed to Trackpilots via their REST API. Changes to screenshot, idle, and work hour settings may take up to 1 hour to reflect on Trackpilots' EXE (per their documentation).

WHO CAN ACCESS SETTINGS

Only Client Admin role can access the Settings module. Client Manager role has NO access to any settings sub-section. This is enforced at both the UI navigation level (menu item hidden) and the API level (403 Forbidden if Manager attempts API call).

10.1 Manage Team

Field/Column	Description
Team Name	Team name displayed in list
Members Count	Number of employees currently assigned to this team
Created On	Date team was created
Edit Action	Opens inline edit modal – change team name only
Delete Action	Soft delete – team removed, employees become unassigned; their historical data is retained

- 'Create New Team' button → modal with Team Name input + Create button
- Team names must be unique within the organization
- Trackpilots API: GET /v1/manage-teams/fetch-all-teams | POST create-team | PATCH update-team | DEL delete-team

10.2 Manage Employee

Column	Description
Employee Name + Email	Full name with email below and avatar initials
Role	Assigned access role badge (Admin Manager Employee No Access)
Work Mode	Office Remote Hybrid – work environment type
OS	Operating system from last webhook event: Windows macOS Linux
EXE Version	Trackpilots agent version installed on employee's machine
Last Login	Timestamp of last webhook event received from this employee
Actions	Edit (modal) Delete (soft delete with confirmation)

- 'Add a New Employee' button → invite flow: enter name, email, select team, select role → Trackpilots invite email sent via POST /v1/manage-employees/add-employee-via-invite-link
- Edit Employee modal: change name, team assignment, role, work mode
- Delete: soft delete – employee.status = inactive; their historical data is preserved
- Trackpilots API: GET /v1/manage-employees/fetch-all-employees | POST add | PATCH update | DEL delete

10.3 Work Day Settings

Configure which days of the week are working days for each employee individually.

Column	Description
Employee Name + Email	One row per employee in the organization
Monday → Sunday	One checkbox per day — checked = working day, unchecked = day off for this employee
Save (per row)	Saves configuration immediately for that employee; effective within 1 hour

- Affects attendance reporting: absent flags are only applied on checked working days
- Trackpilots API: GET /v1/settings/fetch-work-day-settings | PATCH /v1/settings/update-work-day-settings

10.4 Expected Work Hours

Column	Description	Default
Employee Name + Email	One row per employee	—
Expected Work Hours Per Day	Dropdown: 4h 5h 6h 7h 8h 9h 10h	8 hours
Expected Productive Work Hours Per Day	Dropdown: same options. Must be ≤ Expected Work Hours	6 hours
Expected In Time Everyday	Time picker HH:MM AM/PM — configured expected daily start time	08:00 AM
Save	Per-row save button — saves immediately	—

- These values drive: Productivity target calculation | Late check-in detection | Overtime calculation | Progress bar on dashboard
- Trackpilots API: GET /v1/settings/fetch-expected-work-hours-settings | PATCH update

10.5 Screenshot Settings

Column	Description	Default
Employee Name + Email	One row per employee	—

Column	Description	Default
Screen Capture On/Off	Toggle — enable or disable screenshot capture for this employee	ON
Blur Screenshot	Toggle — applies Gaussian blur server-side when Soniq stores the image (employee privacy)	OFF
Screen Capture Interval (Mins)	Numeric stepper 1-60 — frequency of screenshot capture	1 minute
Save	Per-row save button	—

- When Screen Capture is OFF: Soniq backend ignores screenshot webhook events for this employee (does not store or display)
- When Blur is ON: Sharp library applies Gaussian blur to the imageBuffer before uploading to S3; original is never stored
- Trackpilots API: GET /v1/settings/fetch-screenshot-settings | PATCH update-screenshot-settings

10.6 Idle Alert Settings

Column	Description	Default
Employee Name + Email	One row per employee	—
Idle Alert On/Off	Toggle — enable or disable idle detection for this employee	ON
Minimum Idle Time (Minutes)	Numeric stepper — threshold minutes of no activity before marking as idle. Max 5 minutes as per frozen scope.	5 minutes
Save	Per-row save button	—

- Trackpilots API: GET /v1/settings/fetch-idle-alert-settings | PATCH update-idle-alert-settings

10.7 Stealth Monitoring

Column	Description
Employee Name + Email	One row per employee
Enable Stealth Monitoring	Checkbox — when enabled, monitoring runs in background; Trackpilots system tray icon is hidden from employee

- Legal consent banner must be acknowledged by Client Admin before stealth monitoring can be enabled for ANY employee
- Soniq displays a mandatory warning: 'Stealth monitoring may be subject to local labor laws. Ensure compliance with your jurisdiction before enabling. Soniq is not responsible for legal misuse.'
- Trackpilots API: GET /v1/settings/fetch-stealth-monitoring-settings | PATCH update

10.8 Default Settings

Global defaults that apply to all new employees added to the organization. Per-employee settings (from sections above) always override these defaults.

Section	Fields
Workday Settings	Default working days (Monday–Sunday checkboxes applied to all new employees)
Screenshot Settings	Enable Screenshot default Enable Blur default Screen Capture Interval default
Idle Alert Settings	Enable Idle Alert default Minimum Idle Time default
Work Hour Settings	Expected Work Hours Per Day Expected Productive Work Hours Per Day Expected In Time
Timezone Settings	Organization Timezone — dropdown (UTC IST EST PST GMT etc.)
Stealth Monitoring	Enable Stealth Monitoring default (requires consent banner acknowledge)

- Trackpilots API: GET /v1/settings/fetch-default-settings | PATCH update-default-settings

10.9 Access Control (Roles & Permissions)

Roles List View

Column	Description
Role Name	Role name with avatar letter badge (A = Admin, M = Manager, E = Employee, NA = No Access)
Page Permissions	Accordion list of modules this role can access — expandable to show read/write level per module
Actions	Edit Delete (pre-built roles Admin and No Access cannot be deleted)

Pre-Built Default Roles

Role	Module Access	Configurable?
Admin	Dashboard Teams Screenshots Attendance Reports Settings Access Control — Full access to everything	Read-only — cannot be deleted or restricted
Manager	Dashboard Teams Screenshots Attendance (read-only, own teams only)	Configurable — can be extended or restricted
Employee	No portal access in Phase 1	No portal login
No Access	No pages — user exists but cannot log in to Soniq Client Portal	Cannot be modified

Create New Role

- 'Create a New Role' button → Opens modal
- Enter Role Name (unique within org)
- Module selector: Checkbox list — Dashboard | Teams | Screenshots | Attendance | Reports | Settings | Access Control
- For each module: [No Access] [Read Only] [Full Access] radio options
- Save → Role available in employee role assignment dropdown
- Trackpilots API: GET /v1/access-control/fetch-access-role | POST create | PATCH update | DEL delete

11. Backend Technical Specification

11.1 Webhook Receiver Architecture

Component	Detail
Endpoint	POST https://api.soniq.io/webhooks/trackpilots
Body Parser	express.raw({ type: 'application/json', limit: '50mb' }) — raw body required for HMAC verification
HMAC Validation	Compute HMAC-SHA256(x-webhook-timestamp + '.' + rawBody, WEBHOOK_SECRET). Compare with x-webhook-signature header. Reject with 401 if mismatch.
Timestamp Validation	Reject events older than 5 minutes (prevent replay attacks): $\text{Math.abs}(\text{Date.now}() - \text{timestamp}) > 300000 \rightarrow \text{reject}$
Response Requirement	Must return HTTP 200 within 5 seconds. Process async — push to BullMQ queue immediately, return 200, process in background worker.
Queue	BullMQ job queue backed by Redis. Three queue types: activity-tracking-queue app-tracking-queue screenshot-tracking-queue
Workers	Separate BullMQ worker processes for each queue type. Screenshot worker handles heavy image processing.
Retry Policy	BullMQ: 3 retries with exponential backoff (1s, 5s, 30s). Failed jobs → dead-letter queue for manual review.
Idempotency	Use trackpilots_screenshot_id / tracking.trackingId as unique keys. Check DB before insert to prevent duplicate processing on retry.

11.2 API Endpoint Design

Module	Method	Endpoint	Auth Required	Description
Auth	POST	/api/auth/login	None	Client Portal login — returns JWT + refresh token
Auth	POST	/api/auth/refresh	Refresh token	Get new access token
Auth	POST	/api/auth/logout	JWT	Invalidate refresh token

Module	Method	Endpoint	Auth Required	Description
Dashboard	GET	/api/dashboard/stats	JWT	4 summary stat cards
Dashboard	GET	/api/dashboard/work-summary	JWT	Rainbow chart data
Dashboard	GET	/api/dashboard/live-status	JWT	SSE stream for live scroller
Dashboard	GET	/api/dashboard/top-productive	JWT	Top 5 productive (type=employee website app)
Dashboard	GET	/api/dashboard/top-unproductive	JWT	Top 5 unproductive
Dashboard	GET	/api/dashboard/today-activity	JWT	Today's activity table
Dashboard	GET	/api/dashboard/work-hour-insight	JWT	Bar chart data
Teams	GET	/api/teams	JWT	Team list with live status counts
Teams	GET	/api/teams/:teamId	JWT	Team detail + employee list
Teams	GET	/api/teams/:teamId/calendar	JWT	Calendar heat map data
Employees	GET	/api/employees/:employeeId	JWT	Employee detail page data
Employees	GET	/api/employees/:employeeId/overview	JWT	Overview tab: productivity summary + time tracker
Employees	GET	/api/employees/:employeeId/apps	JWT	App/website usage tab
Screenshots	GET	/api/screenshots	JWT	Screenshot list with filters + pagination
Attendance	GET	/api/attendance	JWT	Attendance list view
Attendance	GET	/api/attendance/timeline	JWT	Timeline view data by hour
Reports	GET	/api/reports/productivity	JWT	Productivity Insights report
Reports	GET	/api/reports/effort	JWT	Effort Utilization report
Reports	GET	/api/reports/attendance	JWT	Attendance report
Reports	GET	/api/reports/app-usage	JWT	App/website usage report

Module	Method	Endpoint	Auth Required	Description
Reports	GET	/api/reports/timesheet	JWT	Timesheet report
Reports	GET	/api/reports/export	JWT	CSV/Excel export for any report
Settings	GET/PATCH	/api/settings/teams	JWT (Admin)	Manage teams CRUD
Settings	GET/PATCH	/api/settings/employees	JWT (Admin)	Manage employees CRUD
Settings	GET/PATCH	/api/settings/work-day	JWT (Admin)	Work day configuration
Settings	GET/PATCH	/api/settings/work-hours	JWT (Admin)	Expected work hours
Settings	GET/PATCH	/api/settings/screenshot	JWT (Admin)	Screenshot settings
Settings	GET/PATCH	/api/settings/idle-alert	JWT (Admin)	Idle alert settings
Settings	GET/PATCH	/api/settings/stealth	JWT (Admin)	Stealth monitoring settings
Settings	GET/PATCH	/api/settings/defaults	JWT (Admin)	Default settings
Roles	GET/POST/PATCH/DELETE	/api/settings/roles	JWT (Admin)	Access control roles CRUD

11.3 Database Schema — Complete Table List

Table	Purpose	Key Columns
organizations	One row per client organization	id (UUID PK), name, trackpilots_org_id, trackpilots_api_key (encrypted), trackpilots_webhook_secret (encrypted), plan_id, timezone, status, created_at
subscriptions	Billing plan per organization	id, org_id (FK), plan_name, monthly_amount, billing_cycle, started_at, expires_at, status

Table	Purpose	Key Columns
teams	Teams within an organization	id (UUID PK), org_id (FK), trackpilots_team_id, name, created_at
employees	Employees within an organization	id (UUID PK), org_id (FK), team_id (FK), trackpilots_user_id, name, email, role_id (FK), profile_pic_url, status, os, app_version, last_seen_at, work_mode, designation, department
roles	Access roles per organization	id (UUID PK), org_id (FK), name, permissions (JSONB: {module: 'dashboard', level: 'read'}[]), is_default (bool)
activity_events	Raw webhook event storage – activity tracking	id, org_id, team_id, employee_id, event_type ENUM(activity app screenshot), work_mode (bool), app_name, app_type, app_category, app_domain, app_full_url, productivity_status, duration_seconds, start_time, end_time, tracking_mode, operating_system, raw_payload (JSONB), received_at
screenshots	Screenshot metadata (images in S3)	id, org_id, team_id, employee_id, trackpilots_screenshot_id, image_url (S3/CDN), is_blurred, app_name, app_type, app_category, app_domain, app_full_url, productivity_status, is_idle, captured_at, operating_system, work_type
daily_summaries	Aggregated per-employee per-day metrics	id, org_id, team_id, employee_id, summary_date, first_checkin, last_checkout, total_work_seconds, productive_seconds, unproductive_seconds, neutral_seconds, idle_seconds, productivity_score (DECIMAL), is_present, is_late, overtime_seconds, screenshots_count
webhook_logs	Log of every webhook received	id, org_id, event_type, signature_valid (bool), processing_status ENUM(queued processed failed), error_message, received_at, processed_at, latency_ms
audit_logs	Admin action log	id, actor_id, actor_type ENUM(super_admin client_admin), org_id, action, target_type, target_id, before_value (JSONB), after_value (JSONB), created_at

Table	Purpose	Key Columns
expected_work_hours_settings	Per-employee work hour config	id, org_id, employee_id, expected_work_hours_per_day, expected_productive_hours_per_day, expected_in_time, updated_at
work_day_settings	Per-employee working day config	id, org_id, employee_id, monday through sunday (booleans), updated_at
screenshot_settings	Per-employee screenshot config	id, org_id, employee_id, screen_capture_enabled, blur_enabled, capture_interval_minutes, updated_at
idle_alert_settings	Per-employee idle alert config	id, org_id, employee_id, idle_alert_enabled, min_idle_time_minutes, updated_at

12. Development Timeline — Target: End of May 2026

TARGET COMPLETION DATE

All Phase 1 modules — including both Super Admin Portal and Client Portal — must be fully developed, tested, and production-ready by end of May 2026. This includes QA testing, UAT with pilot clients, and infrastructure deployment.

Week	Phase	Deliverables	Status
Week 1-2 (Apr 9-20)	Pre-Development Setup	✓ Resolve all Trackpilots dependency items ✓ Purchase server, S3, DB, Redis ✓ Set up domains and SSL ✓ Set up Resend email ✓ Create DB schema ✓ Dev environments configured ✓ GitHub repo + CI/CD pipeline	IN PROGRESS
Week 3-4 (Apr 21 - May 4)	Backend Core	Webhook receiver with HMAC validation BullMQ queues for 3 event types Activity + App event processors Screenshot processor with S3 upload + blur daily_summaries aggregation cron job Employee/org mapping pipeline JWT auth for both portals	PLANNED
Week 5-7 (May 5-18)	Backend APIs + Frontend Start	All REST API endpoints for both portals Client Portal: Dashboard + Teams modules Super Admin Portal: Org management + Employee overview SSE live status endpoint Real-time live scroller	PLANNED

Week	Phase	Deliverables	Status
Week 7-9 (May 12-25)	Frontend — All Modules	Client Portal: Screenshots + Attendance + All 5 Reports + Settings (all 8 sub-sections) + Access Control Super Admin Portal: All tabs + Billing + Audit log + Platform health	PLANNED
Week 9-10 (May 19-28)	QA + Fixes + UAT	Full end-to-end testing Performance testing (webhook load) Security audit (HMAC, JWT, SQL injection, XSS) UAT with 2-3 pilot client organizations Bug fixing Data accuracy validation	PLANNED
Week 10 (May 29-31)	Production Launch	Production deployment DNS cutover Monitoring dashboards live First client onboarded on production Documentation complete	TARGET

13. Out of Scope — Phase 1

Feature	Reason
Employee Self-Service Portal (login)	No employee login. Phase 2.
Mobile Apps (iOS / Android)	Phase 2 — Scale phase.
SonIQ Proprietary EXE Agent	Future roadmap — replace Trackpilots EXE with Soniq-built agent.
Payroll / HR Integration	Not a platform core feature. Permanently out of scope.
Meetings Tab	Placeholder exists in UI; functionality Phase 2.
Regularization / Leave Requests	Attendance corrections deferred to Phase 2.
Announcement Module	Phase 2 communication features.
AI / ML Predictive Analytics	Advanced intelligence features — Phase 3.
Custom Report Builder	Phase 2 — let clients build their own report columns.
Multi-Language Support	English only in Phase 1.
Dark Mode	Phase 2 UI enhancement.
Site Search Report	Optional — only if time permits after 5 core reports.
API Access for Clients	Phase 2 — public REST API for client integrations.

Feature	Reason
White-Label / Reseller Features	Enterprise phase — custom branding per client.
Offline Mode / PWA	Phase 2.
SSO / SAML Authentication	Enterprise phase.
Two-Factor Authentication (2FA)	Phase 2 security enhancement.

14. Appendix

14.1 Glossary

Term	Definition
SonIQ	The product — Employee Productivity & Intelligence Platform (B2B SaaS)
Trackpilots	Third-party vendor providing the desktop EXE agent and webhook data pipeline
Super Admin Portal	Separate internal web app (admin.soniq.io) for Soniq's own team to manage all clients
Client Portal	Separate web app (app.soniq.io) for client organization Admins and Managers
Webhook	HTTP POST request automatically sent by Trackpilots to Soniq when a desktop event is captured
EXE Agent	Trackpilots desktop software installed on employee machines that captures and sends data
HMAC SHA256	Cryptographic signature algorithm used to verify webhook authenticity
BullMQ	Node.js job queue library used for async webhook event processing
daily_summaries	Soniq's aggregated per-employee per-day DB table — backbone of all reports and dashboard widgets
productive_seconds	Sum of app/website usage durations where productivityStatus = 'productive'
idle_seconds	Computed: total_work_seconds minus (productive + unproductive + neutral) seconds
productivity_score	$\text{productive_seconds} / (\text{productive} + \text{unproductive} + \text{neutral}) \times 100$ — scale 0-100

Term	Definition
Work Mode	Employee's EXE is running and actively tracking (workMode=true in webhook)
Privacy Mode	Employee paused tracking (EXE running but workMode=false)
Offline	No webhook event received from employee in the last 5 minutes — computed by Soniq
App Not Installed	Employee exists in Soniq but has never sent a webhook event
S3	AWS Simple Storage Service — used to store Soniq screenshot images
CloudFront	AWS CDN — serves screenshot images fast globally with signed URL expiry
SSE	Server-Sent Events — used for real-time live status feed on dashboard
Multi-tenant	Single Soniq platform instance serving multiple isolated client organizations
Rainbow Chart	Soniq's semicircle donut work summary visualization with 5 color-coded segments

14.2 Complete Trackpilots API Reference

Module	Method	Endpoint	Used For in Soniq
Dashboard	GET	/v1/dashboard/employee-recent-activity	Live employee status scroller
Dashboard	POST	/v1/dashboard/fetch-work-summary	Rainbow semicircle chart
Dashboard	POST	/v1/dashboard/fetch-top-productive-performer	Top Productive panel
Dashboard	POST	/v1/dashboard/fetch-recent-screenshot	Recent screenshots strip
Dashboard	POST	/v1/dashboard/fetch-work-time-analysis	Work Hour Insight chart
Dashboard	POST	/v1/dashboard/fetch-productivity-analysis	Productivity breakdown
Dashboard	POST	/v1/dashboard/fetch-apps-and-websites	App/website usage
Dashboard	POST	/v1/dashboard/fetch-weekly-summary	Weekly summary view

Module	Method	Endpoint	Used For in Soniq
Dashboard	POST	/v1/dashboard/fetch-work-time-efficiency	Efficiency metrics
Teams	POST	/v1/my-team/fetch-my-team	Team member list
Teams	GET	/v1/my-team/fetch-time-logs	Time logs per employee
Teams	GET	/v1/my-team/fetch-work-summary	Team work summary
Teams	GET	/v1/my-team/fetch-screenshot	Team screenshot list
Teams	GET	/v1/my-team/fetch-apps-and-websites-usage	Team app usage
Insights	POST	/v1/insights/fetch-productivity-analysis	Productivity Insights report
Screenshots	POST	/v1/proof-of-work/fetch-screenshot	Screenshot module
Timesheet	POST	/v1/proof-of-work/fetch-timesheet	Timesheet data
App Usage	POST	/v1/pow--app-usage/fetch-work-summary	App usage work summary
App Usage	GET	/v1/pow--app-usage/fetch-websites-usage	Website usage list
App Usage	GET	/v1/pow--app-usage/fetch-apps-usage	App usage list
App Usage	POST	/v1/pow--app-usage/fetch-timeline	Timeline view data
Reports	POST	/v1/report-analysis/fetch-monthly-summary	Monthly overview report
Reports	POST	/v1/report-analysis/fetch-attendance-summary	Attendance summary report
Reports	POST	/v1/report-analysis/fetch-delayed-check-in	Delayed check-in analysis
Reports	POST	/v1/report-analysis/fetch-extra-hours-summary	Extra hours report
Employees	GET	/v1/manage-employees/fetch-all-employees	Manage Employee list
Employees	POST	/v1/manage-employees/add-employee	Add employee manually
Employees	POST	/v1/manage-employees/add-employee-via-invite-link	Invite employee via email

Module	Method	Endpoint	Used For in Soniq
Employees	PATCH	/v1/manage-employees/update-employee	Edit employee details
Employees	DEL	/v1/manage-employees/delete-employee	Remove employee
Teams CRUD	GET	/v1/manage-teams/fetch-all-teams	Fetch all teams
Teams CRUD	POST	/v1/manage-teams/create-team	Create new team
Teams CRUD	PATCH	/v1/manage-teams/update-team	Rename team
Teams CRUD	DEL	/v1/manage-teams/delete-team	Delete team
Settings	GET/PATCH	/v1/settings/fetch-work-day-settings + update	Work day configuration
Settings	GET/PATCH	/v1/settings/fetch-expected-work-hours-settings + update	Expected work hours
Settings	GET/PATCH	/v1/settings/fetch-screenshot-settings + update	Screenshot settings
Settings	GET/PATCH	/v1/settings/fetch-idle-alert-settings + update	Idle alert settings
Settings	GET/PATCH	/v1/settings/fetch-stealth-monitoring-settings + update	Stealth monitoring
Settings	GET/PATCH	/v1/settings/fetch-default-settings + update	Default settings
Access Control	GET/POST/PATCH/DEL	/v1/access-control/ (fetch create update delete)	Role management
Webhooks	HOOK	desktop.activity_tracking.captured	Live work mode changes
Webhooks	HOOK	desktop.app_tracking.captured	App/website usage events
Webhooks	HOOK	desktop.screenshot_tracking.captured	Screenshot image captures

SONIQ — Business Requirements Document

Version 3.0 | Phase 1 Frozen Scope | April 2026

Target Completion: End of May 2026 | STATUS: FROZEN — APPROVED FOR DEVELOPMENT